Lab: Analyzing TCP and UDP Traffic with Wireshark

Estimated time: **30** minutes

Objectives

After completing this lab you'll be able to analyze TCP and UDP network traffic. You'll also be able to demonstrate the skills needed to install, configure, and use an open-source network analysis tool, Wireshark, to perform these tasks.

Tools Needed

- Wireshark, a no-charge open-source network analysis tool, installed on your computer.

Note: Wireshark is available for MacOS as well as for Microsoft Windows. This lab uses the Windows version of this application.

Instructions for Downloading and Installing Wireshark

Before you begin your network analysis lab tasks, you'll need to install and configure Wireshark on your local computer system. Follow these steps to get started:

Downloading Wireshark

1. To download Wireshark, vist the official Wireshark website, https://www.wireshark.org/download.html
2. Select the Windows installer to download the latest stable version of Wireshark.

Installing Wireshark

1. Locate the installer on your machine.
2. Click to installer and follow the on-screen instructions, accepting the default options.
3. During the installation, you might be prompted to install WinPcap or Npcap (necessary for capturing network traffic). If so, choose **Yes** and complete the installation.

Important: If running Wireshark in a virtual environment, do not reboot or restart then environment.

Configuring Wireshark for Network Analysis

If Wireshark is not already running, right click on Wireshark icon and select **Run as Administrator**.

Set Up Your Network Interfaces

1. Upon launching Wireshark, the application displays a list of available network interfaces including wireless and Ethernet networks.
2. Select the interface you want to monitor. Typically, this interface will be your active network connection such as your wireless and Ethernet network.

Configure Capture Filters (Optional)

Before starting a capture, you can set up filters to focus on specific traffic such as `tcp`, `udp`, or specific ports.

To add filters, locate the **Capture Filter** field on the main screen, or after starting the capture, you can enter filters in the **Display Filter** bar.

Testing the Installation

1. Start a new capture by selecting your network interface and clicking the blue shark fin icon (Start Capture).
2. Open a web browser and visit any website to generate traffic.
3. Stop the capture after a few seconds by clicking the red square icon (Stop Capture).
4. You should see the captured packets displayed in Wireshark, confirming that the installation is working correctly.

Saving Your Capture (Optional)

After stopping the capture, you can save the capture file for later analysis by selecting **File** > **Save As…** and selecting a location on your system.

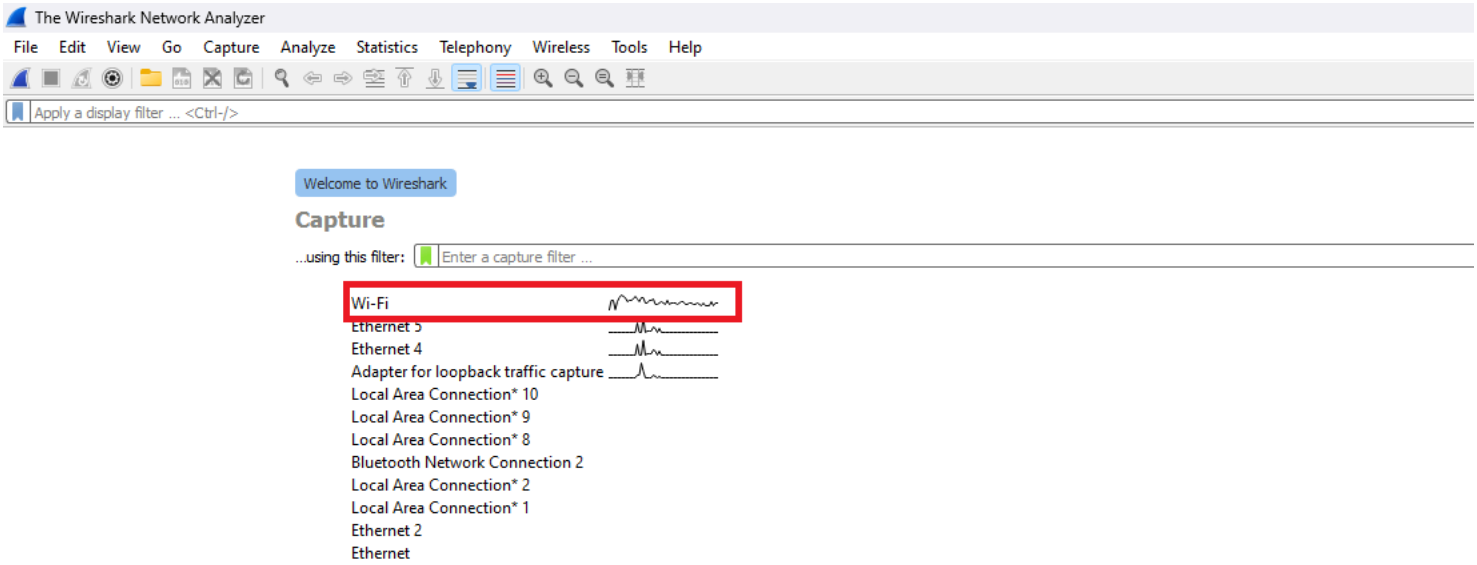
Now that you installed and configure Wireshark, you're ready to begin the lab exercises!

Tasks

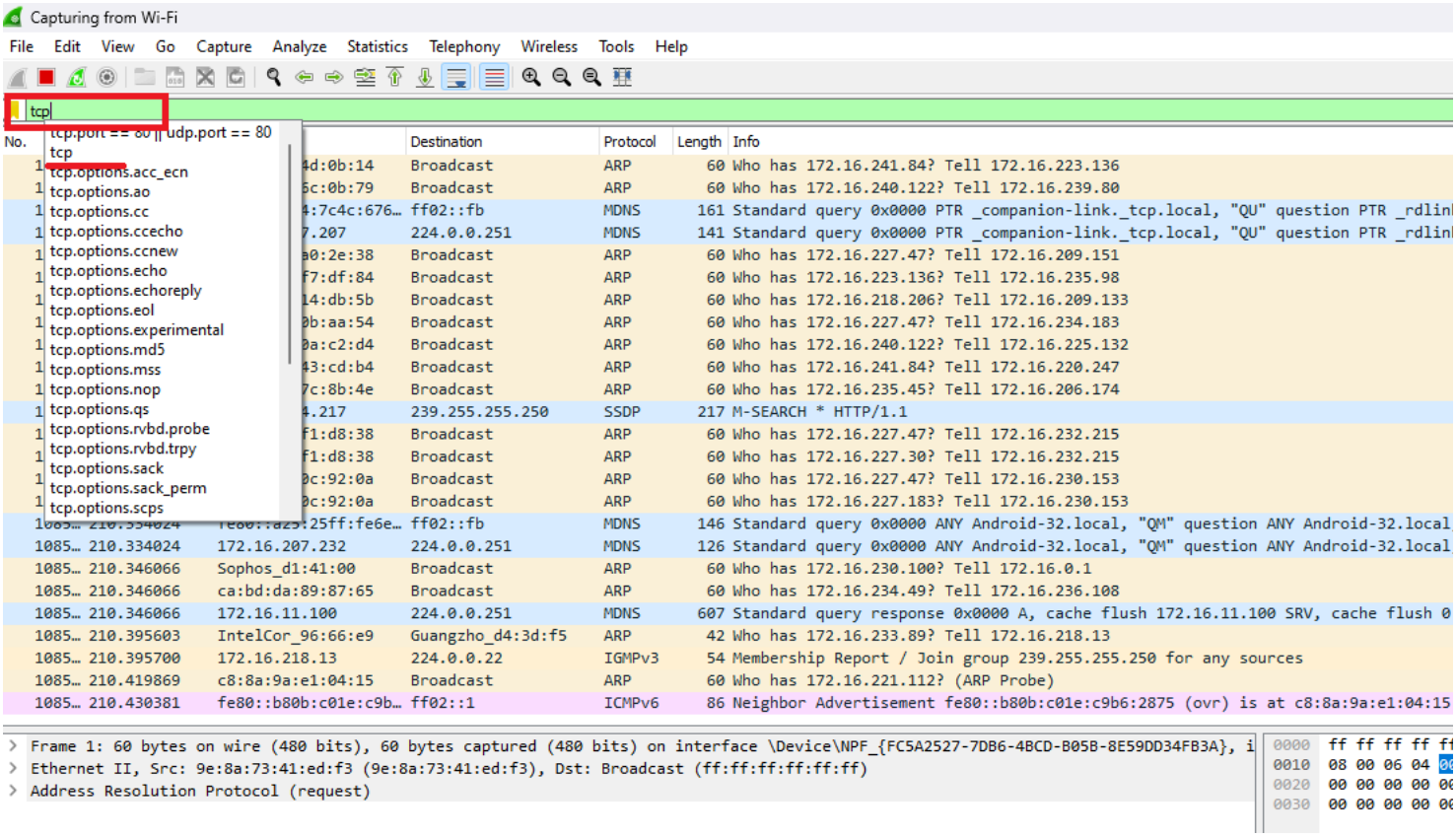
Capture TCP Traffic

Steps:

1. Rick-click the Wireshark icon and select **Run as Administrator** tostart a new capture session.



2. In the filter bar, enter `tcp` and select **Enter** to filter TCP packets.

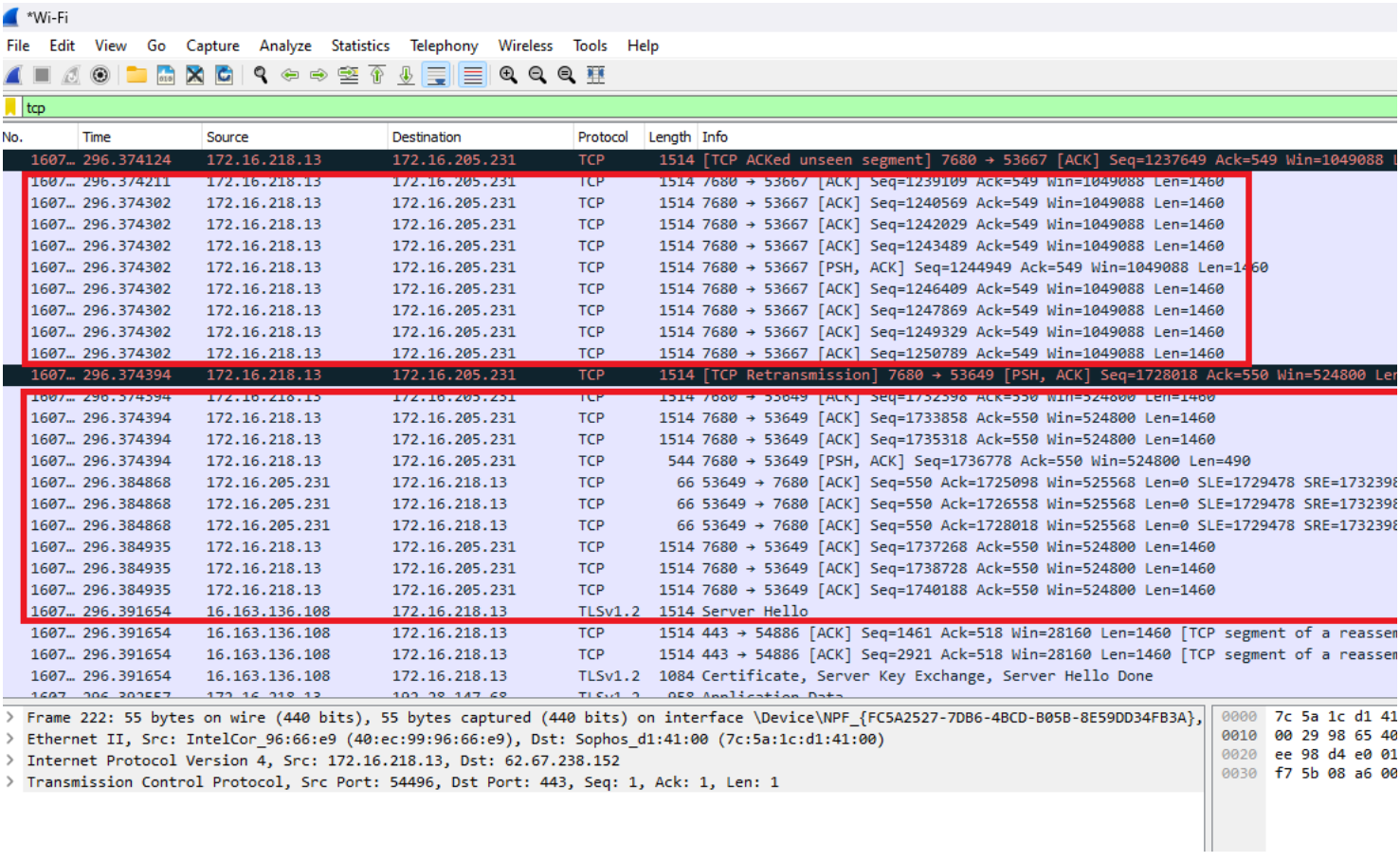


3. Open a web browser and visit a website, such as `https://www.ibm.com/in-en`) to generate TCP traffic.

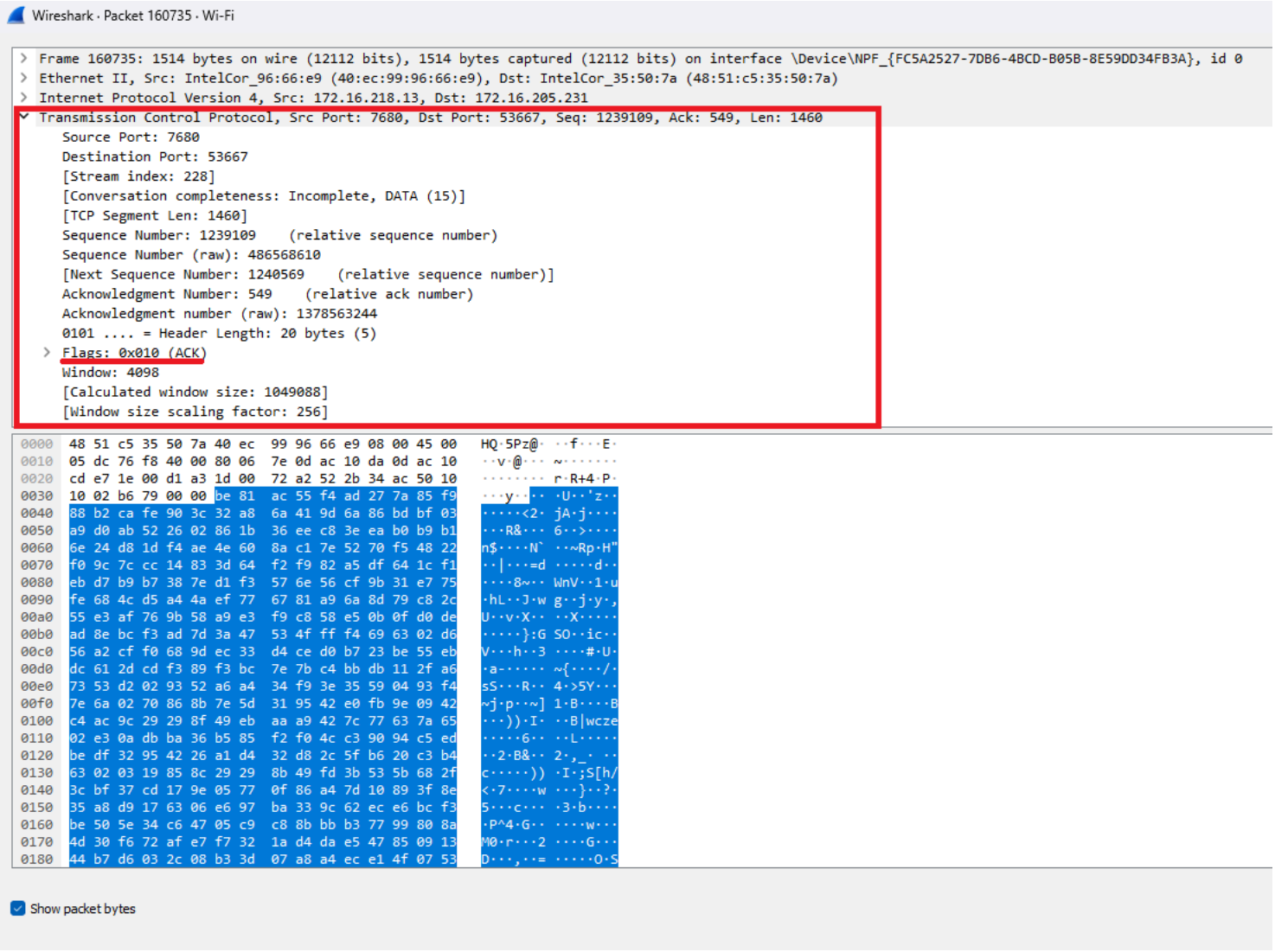
4. Stop the capture after a few seconds.

Capture TCP data

1. Examine the captured packets. Identify packets involved in the TCP three-way handshake including SYN, SYN-ACK, and ACK.

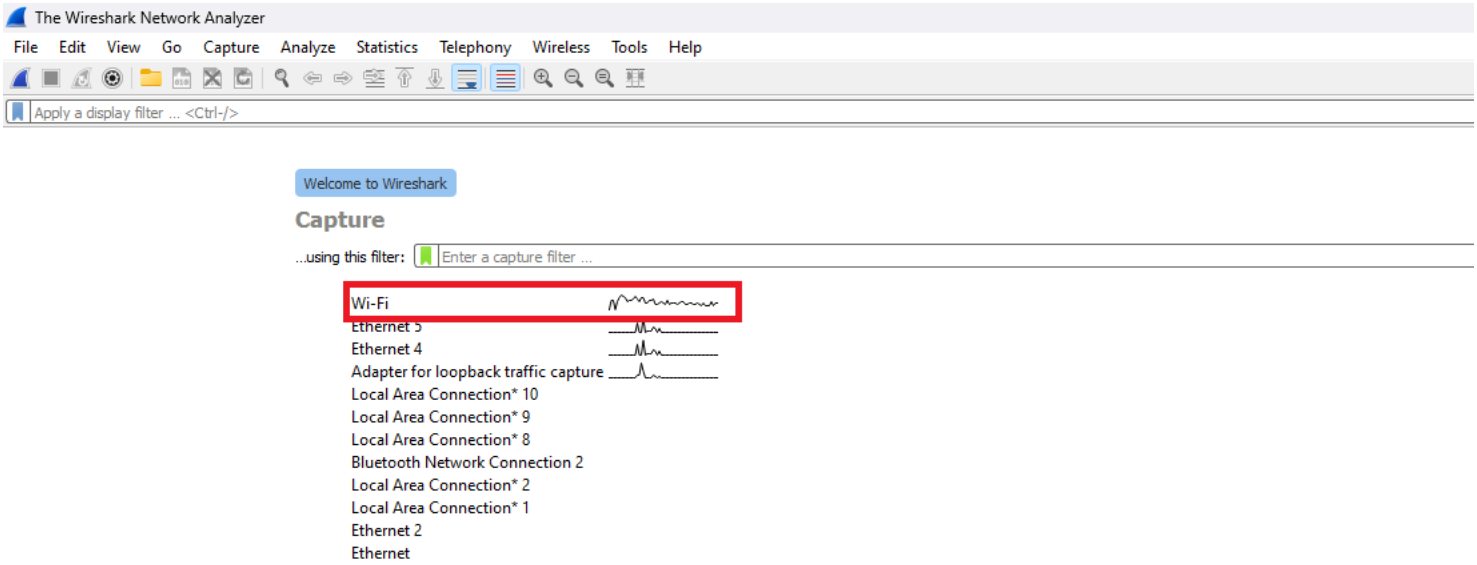


2. Examine the TCP header for sequence numbers, acknowledgments, and flags such as SYN and ACK.

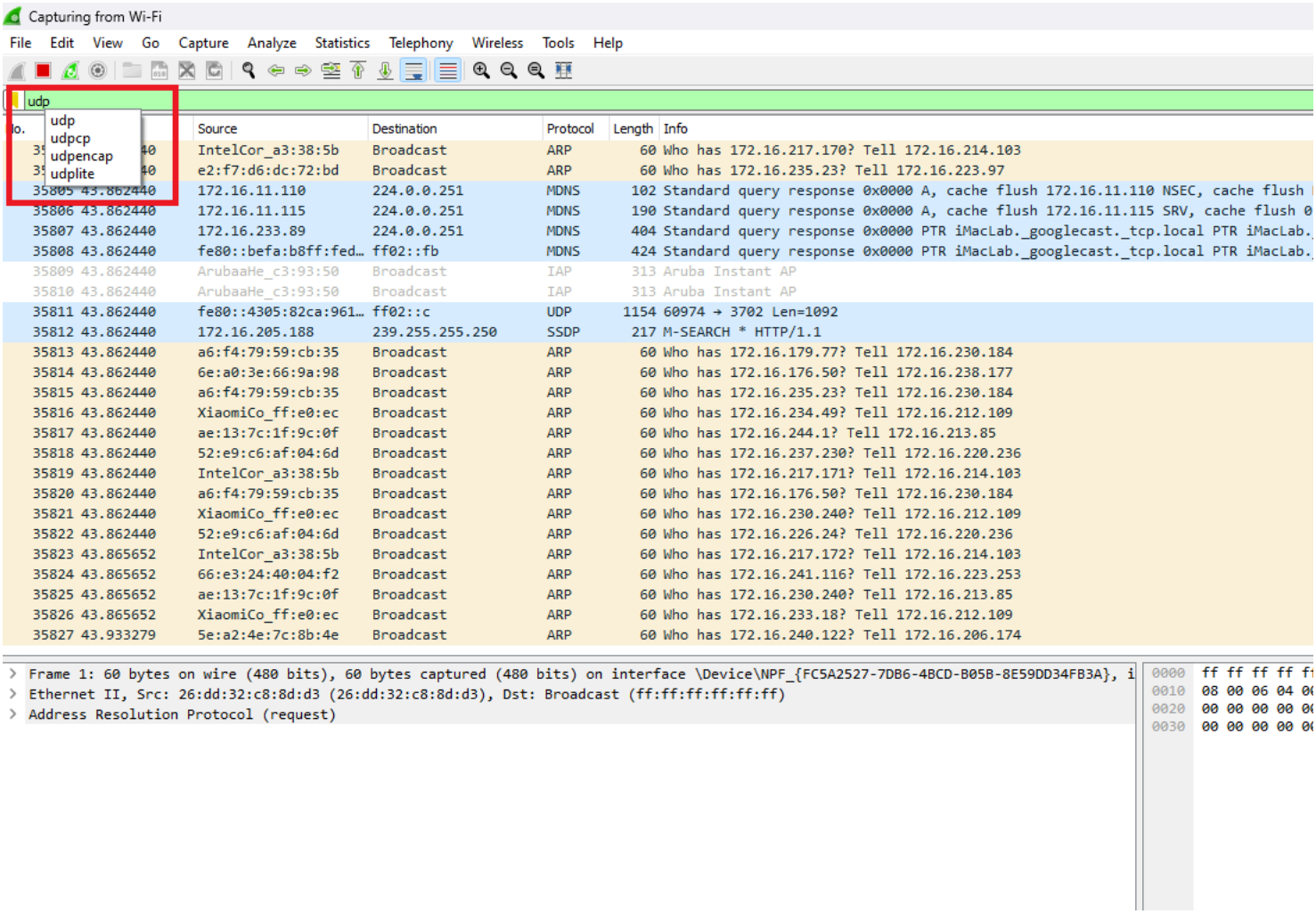


Capture UDP Traffic

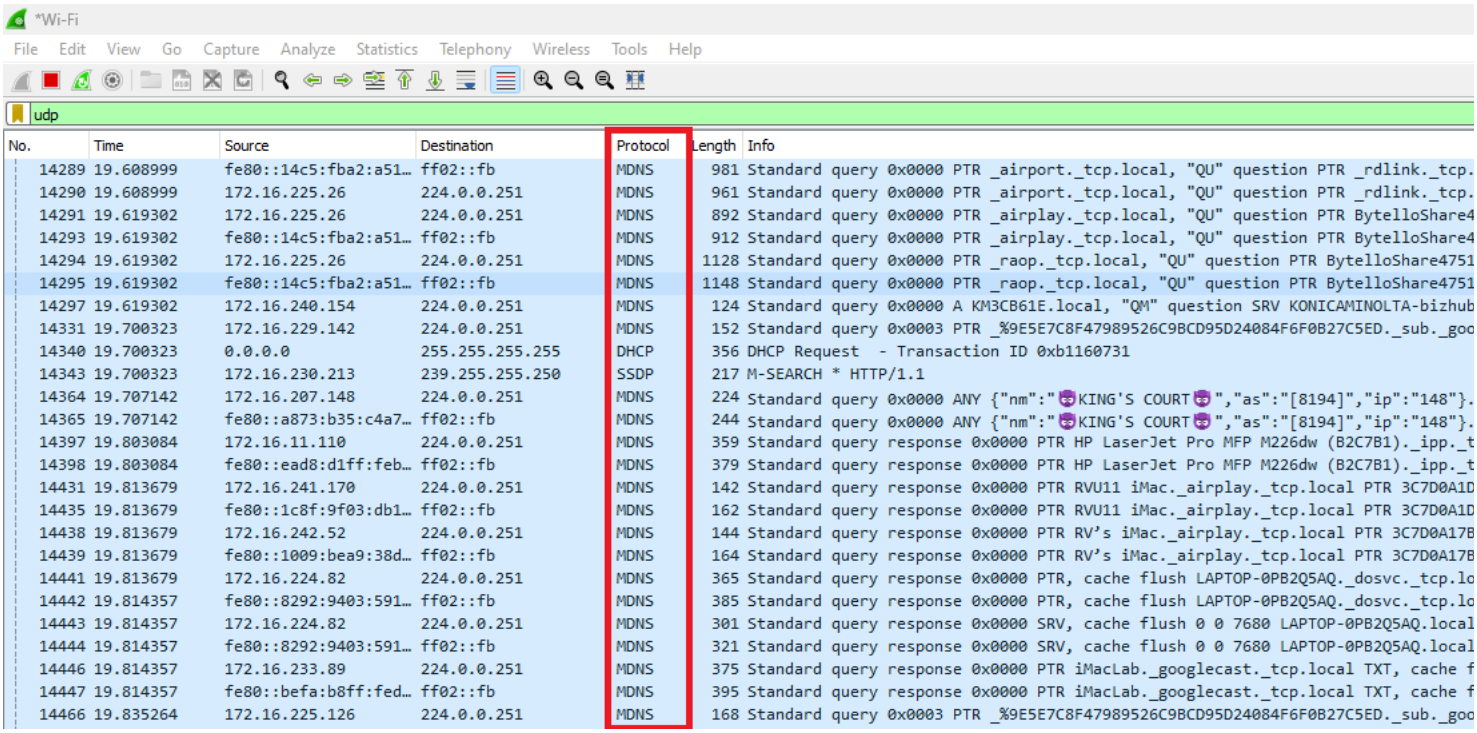
1. In Wireshark, clear the previous capture and start a new session.



2. In the filter bar, enter `udp` and select **Enter** to filter UDP packets.



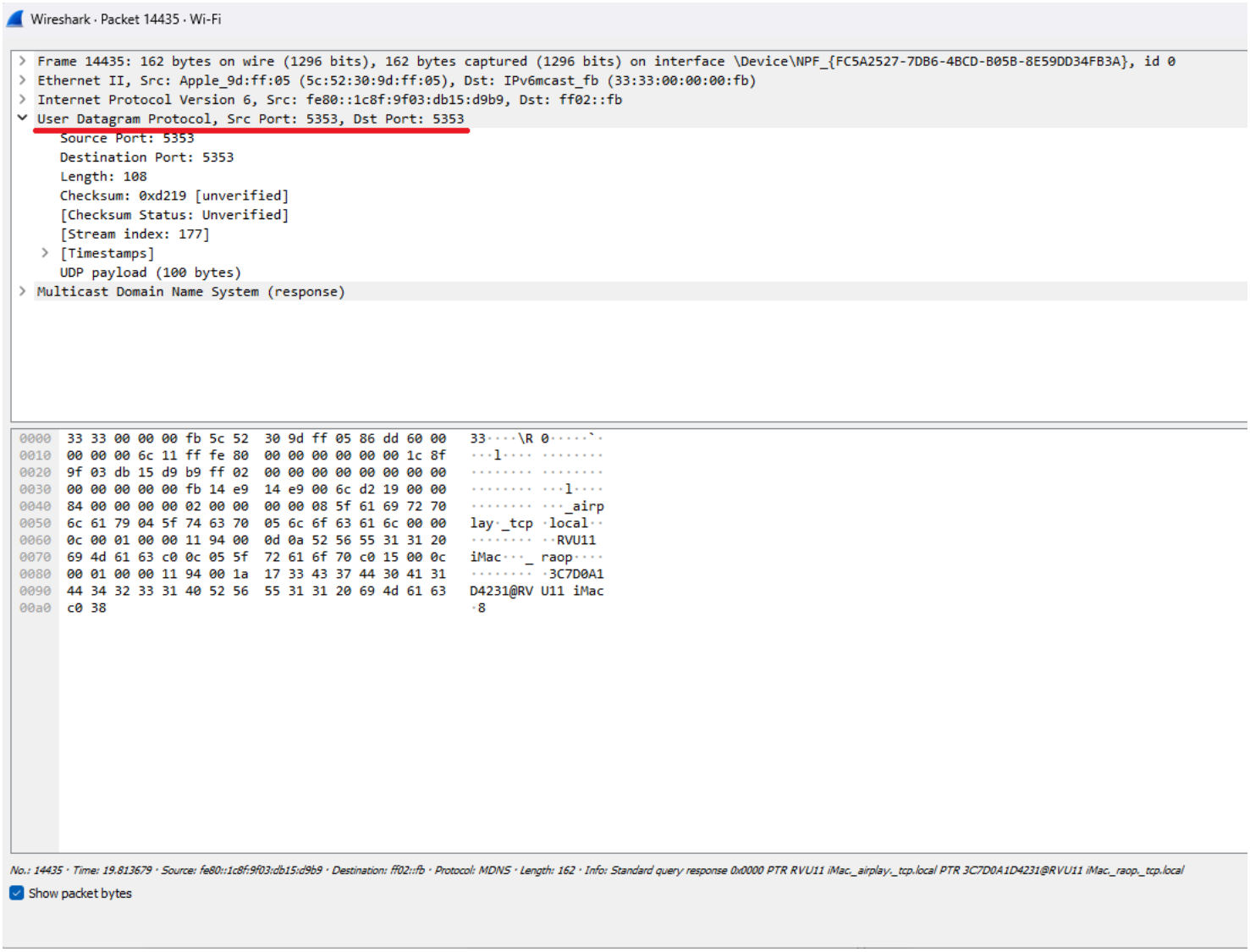
3. Use an application that generates UDP traffic, such as a DNS query or a video streaming service. Allow the service to run for a few seconds.



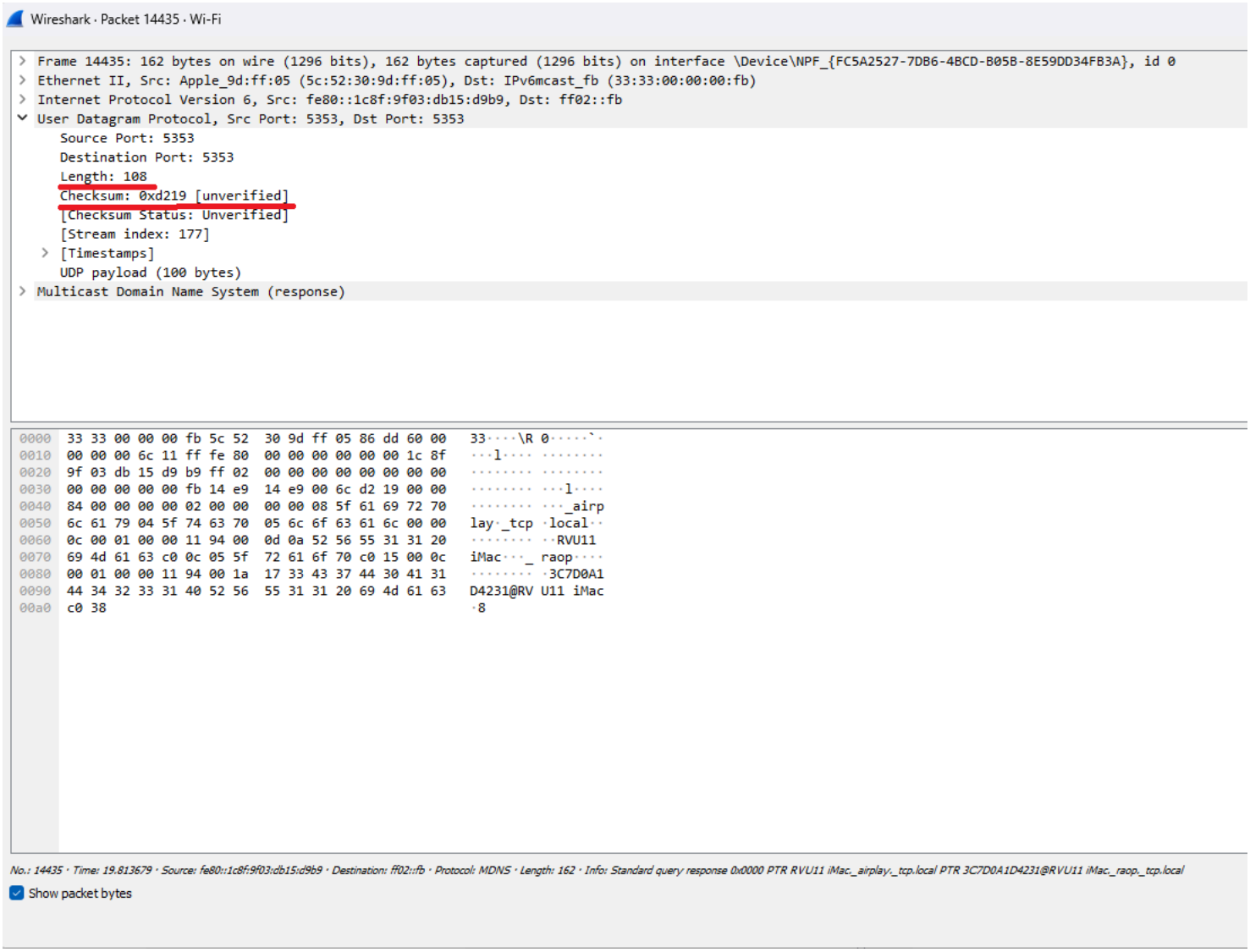
4. Stop the capture after a few seconds.

Analyzing the data

1. Examine the captured packets and identify the source and destination ports.



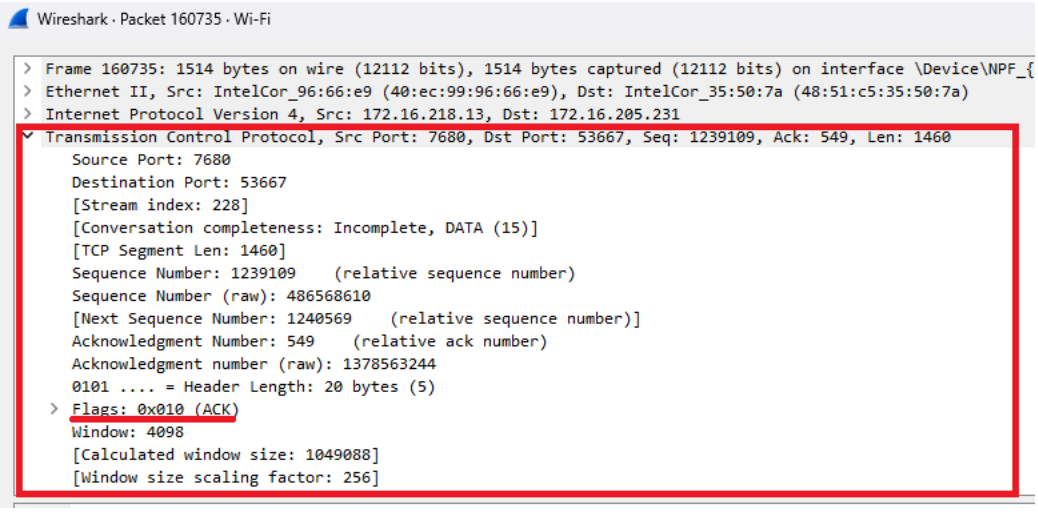
2. Observe the UDP header for the length and checksum fields.



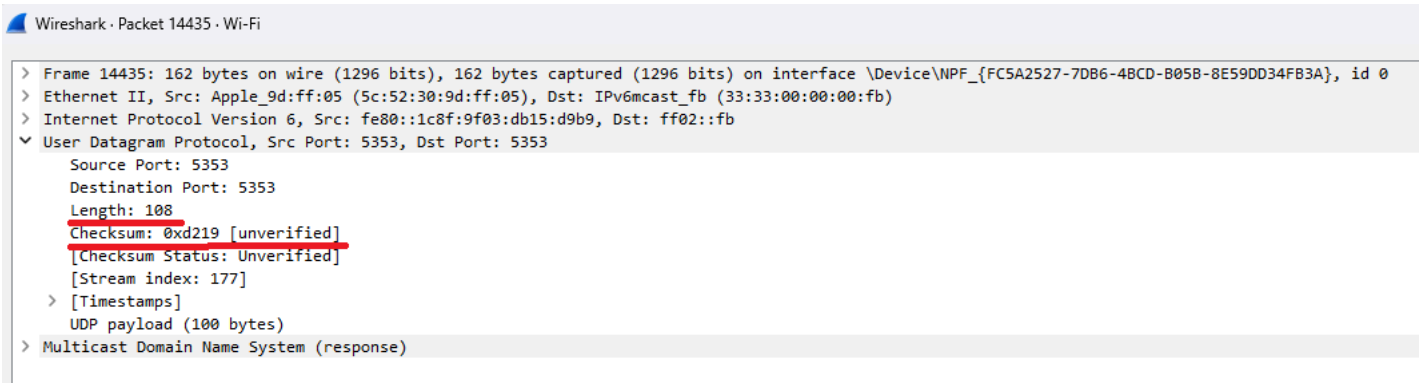
Compare TCP and UDP Packets

1. Compare a TCP packet to a UDP packet captured in previous tasks.

- View a TCP HEADER



- View a UDP HEADER



2. Note the differences in the headers of TCP and UDP packets.

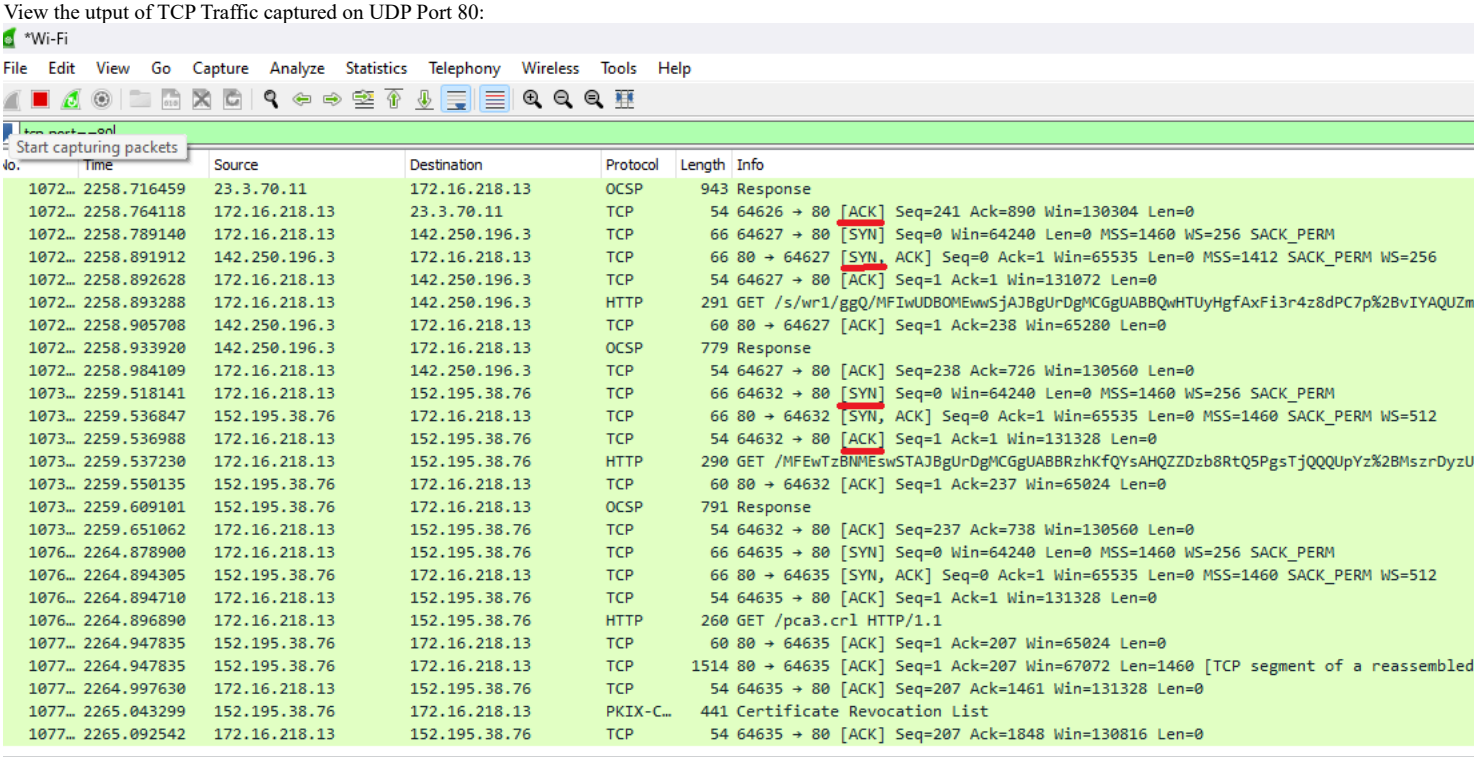
Analyzing the data

- TCP headers include sequence and acknowledgment numbers and flags (SYN, ACK), whereas UDP headers do not.
- UDP packets are generally simpler and have lower overhead compared to TCP packets.

Identify Protocols Using TCP

1. Filter Wireshark captures for a specific TCP port, such as HTTP (port 80) or HTTPS (port 443).
2. Analyze the captured packets to understand the protocol's behavior.

View the utput of TCP Traffic captured on UDP Port 80:



Analysis summary

When analyzing HTTP traffic in Wireshark, start by observing the TCP handshake, where the client and server exchange SYN, SYN-ACK, and ACK packets to establish a connection.

Next, the client sends an HTTP request, specifying the action, suc ash GET or POST, the URL, and relevant headers.

The server then responds with an HTTP response packet that includes a status code, headers, and the requested content. This sequence highlights the structured and reliable nature of HTTP communication over TCP.

Exercises

Exercise 1: Capture and Analyze HTTPS Traffic (TCP)

Objective: Experience how Hypertext Transfer Protocol Secure(HTTPS) operates over TCP by capturing and analyzing its traffic.

1. Open Wireshark using **Run as Administrator** and start a new capture session.
2. In the filter bar, enter `tcp port 443` to capture HTTPS traffic.
3. Open a web browser and visit any secure website (such as `https://www.google.com`).
4. Stop the capture after a few seconds.
5. Analyze the captured packets to identify the TCP three-way handshake (SYN, SYN-ACK, ACK).
6. Observe the encrypted HTTPS traffic and note the absence of visible content due to encryption, while still understanding the flow of requests and responses.

Exercise 2: Capture and Analyze DNS Traffic (UDP)

Objective: Experience how User Datagram Prootcol (UDP) operates by capturing and analyzing its traffic.

1. Open Wireshark using **Run as Administrator** and start a new capture session.
2. In the filter bar, enter `udp port 53` to capture UDP traffic.
3. To generate DNS traffic, type `nslookup` command in command prompt.
4. Stop the capture after a few seconds.
5. Analyze the captured data to look for DNS requests and responses. Examine the UDP headers.

Exercise 3: Capture and Analyze ARP Traffic (Layer 2)

Objective: Experience how Address Resolution Prootcol (ARP) operates by capturing and analyzing its traffic.

1. Open Wireshark using **Run as Administrator** and start a new capture session.
   In the filter bar, enter `ARP` to capture ARP traffic.
2. To enerate ARP traffic, type `arp -a` command in the command prompt.
3. Stop the capture after a few seconds.
4. Analyze the capture lookng for ARP requests and responses. Examine ARP packet details.

Conclusion

This lab provided a hands-on exploration of both TCP and UDP protocols using Wireshark, allowing you to capture and analyze real-world network traffic. By examining HTTPS traffic over TCP, you observed the structured and reliable communication process, including the critical TCP handshake. In contrast, the analysis of SNMP traffic over UDP highlighted the protocol's simplicity and efficiency in connectionless communication. Through these exercises, you gained a deeper understanding of how different protocols leverage the underlying transport layers to meet specific communication needs, reinforcing the fundamental concepts of network traffic analysis.

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